

Multiple Choice (10 marks)

1. Upon the addition of heat to an unknown amount of KCLO_3 , .96 g of oxygen was liberated. How much KCLO_3 was present?
 - (a) 4.5 g
 - (b) 2.2 g
 - (c) 4.0 g
 - (d) 2.0 g
 - (e) 2.5 g
2. Of the following ionic substances, which has the greatest lattice enthalpy?
 - (a) CaS
 - (b) MgS
 - (c) NaCl
 - (d) LiCl
 - (e) NaF
3. The maximum safe concentration of fluoride ion in drinking water is 2.0 ppm(parts per million). Express this concentration in milligram %.
 - (a) 2 mg %
 - (b) 0.0020 mg %
 - (c) 0.002 mg %
 - (d) 0.02 mg %
 - (e) 20 mg %
4. N^{13} decays by β^+ emission. The maximum kinetic energy of the β^+ is 1.19 MeV. What is the nucleidic mass?
 - (a) 13.00573 amu
 - (b) 13.00128 amu
 - (c) 13.00987 amu
 - (d) 13.00335 amu
 - (e) 13.00739 amu
5. One mole of ideal gas at 300 K is expanded adiabatically and reversibly from 20 to 1 atm. What is the final temperature of the gas, assuming $C_V = (3/2) R$ per mole?
 - (a) 135 K

- (b) 150 K
- (c) 75 K
- (d) 110 K
- (e) 180 K

True / False (10 marks)

Answer True or False

6. True or False: For a second-order reaction, halving the half-life indicates that the initial concentration was quadrupled.
7. True or False: The hydronium ion concentration of a 0.1 M acetic acid solution is approximately 0.0013 M.
8. True or False: The magnitude of the spin magnetic moment of an electron is $2.20 \times 10^{-23} J/T$.
8. True or False: Transition metals typically lose their outermost s-block electrons when forming bonds, leading to a common oxidation state of +2.
9. True or False: Increasing the angle away from the Z axis causes the electron density to concentrate on the equatorial plane.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the solid-state structures of the principal allotropes of elemental boron, focusing on the role and characteristics of B_5 pentagons in these structures.
12. Calculate the amount of heat energy required to raise the temperature of 3.0 liters of water from 20°C to 80°C and discuss the underlying principles of heat transfer involved in this process.

End of Paper